

Exercise 2: High-dimensional spaces

Teaching Assistant: Jonas Linkerhäger

Lecturer: Lorenzo Baldassari

Problem 1 (Angle between vectors in high dimension):

Consider two random vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ with i.i.d. Rademacher variables as coordinates (i.e. x_i, y_i take values ± 1 with equal probability). The angle θ between these two vectors is such that

$$\cos \theta = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|}.$$

Show that

$$\Pr \left(|\cos \theta| \geq \sqrt{\frac{2 \log d}{d}} \right) \leq \frac{2}{d}.$$

Problem 2 (Norm of Gaussian vectors):

In the lecture, we have shown that if $\mathbf{x} \in \mathbb{R}^d$ is a random vector with independent Gaussian coordinates X_i such that $\mathbb{E}X_i^2 = 1$, then, for all $t \geq 0$,

$$\Pr(|\|\mathbf{x}\|_2 - \sqrt{d}| \geq t) \leq 2 \exp(-ct^2).$$

1. Show that $\sqrt{d} - C \leq \mathbb{E}\|\mathbf{x}\|_2 \leq \sqrt{d} + C$ where $C > 0$ is a constant.
2. Show that $\text{var}(\|\mathbf{x}\|_2) \leq C'$ where $C' > 0$ is a constant.

Hint: By Stirling's approximation to the Gamma function: $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt \leq z^z$.

Problem 3 (Random projection with Gaussian matrices):

Consider a matrix $\mathbf{A} \in \mathbb{R}^{k \times d}$ with i.i.d. entries $a_{ij} \sim \mathcal{N}(0, 1)$. Let $\mathbf{v} \in \mathbb{R}^d$, then show that there exists $c > 0$ such that for $\epsilon \in (0, 1)$,

$$\Pr(|\|\mathbf{A}\mathbf{v}\| - \sqrt{k}\|\mathbf{v}\|| \geq \epsilon\sqrt{k}\|\mathbf{v}\|) \leq e^{-ck\epsilon^2}.$$

Hint: use the following Gaussian Annulus Theorem proved in class:

Theorem 1 (Gaussian Annulus Theorem). *For a d -dimensional spherical Gaussian of mean 0 and variance 1, for all $\beta \leq \sqrt{d}$,*

$$\Pr(|\|\mathbf{x}\| - \sqrt{d}| \geq \beta) \leq e^{-c\beta^2}.$$